

Stereo Visual Odometry: An ORB-SLAM 3-Inspired Pipeline

CSE 429 – Computer Vision and Pattern Recognition — Spring 2026 — Anas Badr, Solo project

Problem statement

The goal is to estimate, in real time, the 6-DoF trajectory of a moving stereo camera and a sparse 3D point cloud of the surrounding scene from rectified stereo image pairs alone (no IMU, no GPS, no active depth sensor).

Input: a stream of rectified stereo image pairs together with the corresponding intrinsic and extrinsic calibration.

Output: a per-frame camera pose expressed in a global world frame, an accumulated sparse map of triangulated 3D points, and a live 3D visualization (trajectory line, camera frustums, point cloud, current camera image). The system targets driving-style outdoor sequences (KITTI Odometry) and short hand-held captures from a ZED stereo camera.

Approach

The pipeline is feature-based and follows the architectural sketch of ORB-SLAM 2/3, simplified to a frontend-only stereo VO (no IMU integration, no loop closure, no Atlas). For each incoming stereo frame the system: (1) extracts ORB features independently in the left and right images; (2) performs *stereo matching* between left and right (epipolar + disparity gating) to obtain a sparse depth map and metric 3D points in the current camera frame; (3) performs *temporal matching* between the previous left frame and the current left frame to produce 3D-2D correspondences against the previous frame's stereo points; (4) estimates relative motion via 3D-2D **PnP with RANSAC** (`cv::solvePnP`); (5) composes the relative pose into a global world pose; and (6) pushes the new pose, world-frame points, and current image to a Pangolin viewer running on a dedicated render thread.

Implementation is in C++ using OpenCV (features, matching, PnP, image I/O) and Pangolin (3D visualization). The class layout (`Frame`, `Calibration`, `KittiDataset`, `FeatureExtractor`, `StereoMatcher`, `MotionEstimator`, `Trajectory`, `Viewer`) mirrors ORB-SLAM 2's structure at a smaller scale. Gao Xiang's *14 Lectures on Visual SLAM* (slambook) is used as the reference for the underlying multi-view-geometry math.

Experiments and results

Datasets. KITTI Odometry sequence **07** (~1,100 frames, short closed loop) and sequence **00** (~4,500 frames, urban) for quantitative evaluation against ground truth. A supplementary **ZED** stereo capture provides a qualitative non-driving demo; the ZED is accessed as a generic USB stereo camera via OpenCV (`cv::VideoCapture`), and rectification + undistortion are performed manually from the camera's factory intrinsic and extrinsic parameters (no Stereolabs SDK is used).

Existing code reused. OpenCV (ORB extraction, RANSAC PnP, stereo BFMatcher, image I/O), Pangolin (windowing, 3D rendering, GL textures). The slambook examples and the publicly released ORB-SLAM 2 source are used only as architectural references; no code is copied.

Implemented from scratch. KITTI dataset loader, calibration parser, ZED rectification utility built from factory intrinsics + extrinsics (`cv::stereoRectify` → `cv::initUndistortRectifyMap` → `cv::remap`), sparse stereo matcher (with epipolar and disparity filtering), motion estimator (RANSAC PnP wrapper), trajectory I/O in KITTI format, ATE / RPE / drift evaluation, and a multi-threaded Pangolin viewer (trajectory line, recency-coloured camera frustums, FIFO-capped point cloud, side-by-side stereo image strip).

Evaluation metrics. (i) **ATE** – RMSE / mean / max of per-frame position error vs. KITTI ground truth; (ii) **RPE** – relative pose error (translation + rotation) over a 10-frame delta; (iii) **Drift %** – final position error divided by ground-truth trajectory length; (iv) qualitative top-down trajectory plot overlaid on ground truth; (v) qualitative live-viewer demo on ZED data. A **constant-pose baseline** (camera assumed stationary) provides a no-effort floor for ATE.

Planned experiments. (a) Per-sequence ATE / RPE / drift table on KITTI 07 and 00, compared to the baseline. (b) Qualitative top-down trajectory plot per sequence. (c) Sensitivity study on ORB feature count (~1000 / 2000 / 3000) measuring accuracy vs. runtime trade-off. (d) Sensitivity on visualisation depth cap and PnP inlier threshold. (e) Qualitative ZED capture demonstrating real-time operation in an unseen environment.

Success criterion. Final-position drift below **10%** of trajectory length on KITTI 07, and a top-down trajectory on KITTI 00 whose shape is visibly recognisable as the ground-truth path. Qualitatively, the Pangolin viewer should render a coherent point cloud and forward-moving camera on both KITTI and ZED.

Expected outcomes / uncertainty. Drift will accumulate monotonically (no backend bundle adjustment, no loop closure), so KITTI 00's long path is likely to drift visibly even when the trajectory shape is correct. Tracking is expected to be robust on textured urban scenes and weaker on low-texture or strongly dynamic frames. The ZED demo is the largest source of uncertainty: low-texture indoor surfaces (white walls, glossy floors) may starve ORB and force tracking loss; the recording route will be scouted in advance to favour textured paths.